

FINAL TASK FOR ANALISI DEI DATI – 2025/26

1. INTRODUCTION

In molecular biology and clinical bioinformatics, high-throughput technologies (such as microarrays and RNA-sequencing) allow researchers to measure the expression levels of thousands of genes simultaneously from a single tissue sample. This data is vital for precision medicine, such as distinguishing between sub-types of highly aggressive cancers (e.g., Acute Lymphocytic Leukemia vs. Acute Myeloid Leukemia) to tailor chemotherapy protocols.

2. GOALS

Your primary objective is to implement a classification framework that handles high-dimensional collinear features and uncovers a statistically robust set of predictive biomarkers. More precisely, your analysis must explicitly explore and document the following core statistical phenomena.

- *Variable Selection Stability*. Identify the effect of multi-collinearity on sparsity, to compute and map true feature inclusion probabilities.
- *Control of False Discoveries*. Contrast the statistically significant genes against the variables selected by your machine learning models.

Your final models will be considered not only on their binary classification error on the hidden test set but heavily on the *sparsity* and the *reproducibility/stability score* of your selected biomarker subset.

3. DATA

The dataset is derived from a clinical study¹ on leukemia patient stratification, and available at [this URL](#). The dataset collects 72 patients, 5 factors,

- `BM.PB`: BM is from bone marrow, PB (rom peripheral blood,
- `Gender`: F is female, M is male
- `Source`: the hospitals from which the samples originated,
- `tissue.mf`: identifies the combinations of source and gender,
- `cancer`: the two types of Acute Lymphoblastic Leukemia (allB and allT), or Acute Myeloblastic Leukemia (aml)

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¹T.R. Golub and D.K. Slonim and P. Tamayo and C. Huard and M. Gaasenbeek and J.P. Mesirov and H. Coller and M.L. Loh and J.R. Downing and M.A. Caligiuri and C.D. Bloomfield and E.S. Lander, *Molecular Classification of Cancer: Class Discovery and Class Prediction by Gene Expression Monitoring*, Science, 1999.

and the values of 7129 gene probes.

4. SUBMISSION

The results of your own analysis and ideas must be summarised in a report (a PDF file) which explains how you have planned to tackle the problem and the possible strategies you have tried to solve the problem. The emphasis is not on the performances of the final method(s) proposed, but on the way you have dealt with the problem.

You are not only allowed but actually encouraged to read up on the subject, and it is recommended to include a list of references in your report. In order to be complete and fair, you are required to cite *all* sources of research material you have used (books, scientific papers, etc.).

4.1. Fairness. This final assignment is a personal piece of work and must not be done in groups. Discussions with colleagues or experts, although discouraged, should be reported for fairness.

4.2. Use of AI. Use of AI tools, and Large Language Models (LLM) in particular, is in general not allowed. You are welcome to use any tool that is suitable to improve your report, for instance spell checker, grammar suggestions, etc. If you use LLM for editing purposes, please declare it in your report. As the author of your work and of your report, you are responsible for the entire content of the report, which should be correct and *original*. Please avoid, for instance, including reference lists generated by an LLM, but include only those references that are pertinent (and that you have actually read!)

4.3. Report submission. You will upload your report on the *e-learning* website. The deadline is by **July 31, 2026** (but early submissions are appreciated). Do not forget to register to the exam on the page <https://esami.unipi.it> (deadline is **July 24, 2026**, at **11:59am**). This is not mandatory to conclude the exam, but highly recommended and useful for several purposes. If, for some reason, you need to complete the exam before the scheduled date, for instance for degree completion or as an Erasmus student, please contact us as soon as possible.

You should add, at the end of your report, the link to a script (R or `python`, or any other programming language of your choice) containing the implementation of the *final* method(s) proposed, based on the analysis developed. The script must be shared via a *notebook* on [Google Colab](#). Obviously the script must not contain any errors. Please add a link to the notebook in your report.

It is not necessary (and in fact useless) for the script to contain the entire analysis. The recommendation is that the output of your scripts will be a detailed account of your conclusions. The numbers, without any explanation about their meaning, are not really helpful.